Evaluating Spatial Reasoning in Vision-Language Models: A Diagnostic Benchmark with Mechanistic Analysis

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Abstract

We present a systematic benchmark for evaluating spatial reasoning in vision-language models (VLMs), decomposing spatial understanding into interpretable dimensions: egocentric versus allocentric perspective-taking, viewer-centered versus object-relative reference frames, and multiple spatial relation axes. Our benchmark employs automated ground truth generation from scene images using segmentation and depth estimation, enabling exact-match evaluation across 3672 queries. Evaluating three state-of-the-art VLMs, we find consistent performance gaps between egocentric and allocentric reasoning (11–25 percentage points), with allocentric tasks requiring perspective transformation proving substantially harder. Through mechanistic analysis of internal activations and attention patterns, we identify distinct failure mechanisms: egocentric failures correlate with insufficient visual attention (grounding failures), while allocentric failures show no attention deficit (reasoning failures). These findings suggest that improving spatial reasoning requires task-specific interventions rather than uniform approaches.

1 Introduction

2 Related Work

3 Methodology

We present a benchmark for evaluating spatial reasoning in vision-language models that isolates relational reasoning from perceptual confounds. By providing labeled scene visualizations and structured queries with geometrically-computed ground truth, we enable precise diagnosis of model capabilities across spatial reasoning dimensions. Figure 1 illustrates our two-stage pipeline.

3.1 Spatial Graph Generation

Given an input image, we construct a spatial graph $G = (V, E)$ where nodes V represent detected objects and edges E encode pairwise spatial relations. Each node stores object metadata (label, bounding box, segmentation mask, depth), and each edge stores the spatial relation between the connected objects.

Object detection and segmentation. We employ Grounded-SAM, combining Grounding DINO for open-vocabulary detection with SAM for precise mask generation. Given a set of object category

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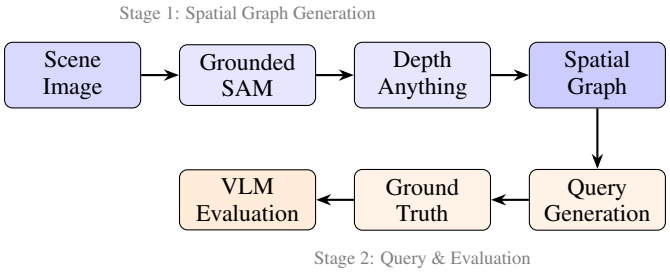

Figure 1: Pipeline overview. Stage 1 constructs a spatial graph encoding object locations and relations. Stage 2 generates queries with geometric ground truth for exact-match evaluation.

prompts, the model returns bounding boxes and segmentation masks for all detected instances. Each object receives a unique identifier by appending an instance index to its category label (e.g., `chair_0`, `chair_1`). We filter detections by confidence threshold and remove heavily overlapping boxes via non-maximum suppression.

Depth estimation. We apply Depth Anything V2 to obtain a dense relative depth map $D \in \mathbb{R}^{H \times W}$, where lower values indicate closer proximity to the camera. For each detected object with segmentation mask M_i, we compute its representative depth as the median depth value within the mask: $d_i = \text{median}(\{D[p] : p \in M_i\})$. The median provides robustness to depth discontinuities at object boundaries.

Spatial relation computation. For each ordered pair of objects (o_i, o_j), we compute spatial relations along four axes by comparing their bounding box centers (c_x, c_y) and depth values d:

- **Horizontal:** o_i is *left* of o_j if $c_x^{(i)} < c_x^{(j)}$, else *right*.
- **Vertical:** o_i is *above* o_j if $c_y^{(i)} < c_y^{(j)}$, else *below*.
- **Depth:** o_i is *in front of* o_j if $d_i < d_j$ (closer to camera), else *behind*.
- **Saliency:** o_i is *foreground* relative to o_j if it is both closer (lower depth) and occupies a larger screen area, else *background*.

Relations are stored as directed edges in the spatial graph, enabling efficient lookup during query generation.

3.2 Query Taxonomy

We organize queries along three orthogonal dimensions that isolate specific reasoning competencies.

Egocentric vs. Allocentric. *Egocentric* queries use the viewer's (camera's) reference frame: "Is the chair to the left or right of the table?" *Allocentric* queries require perspective transformation by specifying an anchor object and facing direction: "Standing at the chair facing the window, is the lamp to your left or right?" This dimension tests mental rotation and reference frame transformation—a capability known to be challenging for both humans and models.

Viewer-centered vs. Object-to-object. *Viewer-centered* queries ask about object positions relative to the observer ("Which object is in the foreground?"), while *object-to-object* queries ask about relations between two specific objects ("Is object A above or below object B?").

Response format. We include two formats: *binary choice* (select between two relation labels, e.g., left vs. right) and *multiple choice* (select among several labels, e.g., left/right/front/behind).

3.3 Ground Truth Computation

Ground truth answers are computed geometrically from the spatial graph, ensuring objective and reproducible evaluation without human annotation.

Egocentric relations. Relations are computed directly in image coordinates. For a query about objects o_i and o_j, we retrieve the precomputed relation from the spatial graph edge (o_i, o_j).

Allocentric relations. We transform the coordinate system to the anchor object's reference frame. Given anchor object A at position $\mathbf{p}_A$ and facing direction toward object F at $\mathbf{p}_F$, we define a local coordinate system with forward vector $\hat{\mathbf{f}} = (\mathbf{p}_F - \mathbf{p}_A)/\|\mathbf{p}_F - \mathbf{p}_A\|$. For a target object T, we compute its position relative to the anchor: $\mathbf{r} = \mathbf{p}_T - \mathbf{p}_A$. Left/right is determined by the sign of $\hat{\mathbf{f}} \times \mathbf{r}$ (cross product), and front/behind by the sign of $\hat{\mathbf{f}} \cdot \mathbf{r}$ (dot product).

3.4 Evaluation Protocol

Input format. Models receive the scene image with overlaid bounding boxes and text labels, plus a natural language query. The visual annotation ensures models can identify referenced objects without requiring object detection capabilities, isolating spatial reasoning from perception.

Metrics. We report accuracy as the fraction of correctly answered queries, with breakdowns by task type (egocentric/allocentric), frame type (viewer-centered/object-to-object), and relation axis (horizontal/vertical/depth) for fine-grained diagnosis of model capabilities.

4 Results

We evaluate three vision-language models on our spatial reasoning benchmark: GPT-5-mini (OpenAI), Claude Haiku 4.5 (Anthropic), and Qwen2.5-VL-7B (Alibaba). Each model receives the annotated scene image and a structured query, responding with a spatial relation label or object identifier.

4.1 Overall Performance

Table 1 presents overall accuracy across all models. All three models achieve above-chance performance, but substantial gaps exist between egocentric and allocentric reasoning tasks.

Table 1: Overall spatial reasoning accuracy by task type. The egocentric-allocentric gap reveals consistent difficulty with perspective transformation across all models.

Model	Egocentric	Allocentric	Gap	Overall
GPT-5-mini	87.5%	81.2%	6.3	**85.0%**
Claude Haiku 4.5	68.8%	56.2%	12.6	63.8%
Qwen2.5-VL-7B	63.9%	52.8%	11.1	59.8%

GPT-5-mini achieves the highest overall accuracy (85.0%), followed by Claude Haiku 4.5 (63.8%) and Qwen2.5-VL-7B (59.8%). Notably, all models exhibit an egocentric-allocentric gap ranging from 6.3 to 12.6 percentage points, confirming that perspective transformation poses a consistent challenge across architectures.

4.2 Performance by Frame Type

Table 2 breaks down performance by frame type (viewer-centered vs. object-to-object). Object-to-object queries require reasoning about relations between two specific objects, which proves more challenging for some models.

GPT-5-mini shows no degradation between frame types, while Claude Haiku 4.5 exhibits a substantial 17.5 percentage point drop on object-to-object queries. This suggests that Claude Haiku 4.5 relies more heavily on absolute position encoding rather than relational reasoning.

Table 2: Accuracy by frame type. Object-to-object queries require explicit relational reasoning between two objects.

Model	Viewer-centered	Object-to-object	Δ
GPT-5-mini	85.0%	85.0%	0.0
Claude Haiku 4.5	72.5%	55.0%	+17.5
Qwen2.5-VL-7B	61.2%	58.4%	+2.8

4.3 Performance by Relation Axis

Table 3 presents accuracy across the four spatial relation axes. Vertical relations (above/below) are generally easiest, while depth-based relations (front/behind) prove most challenging.

Table 3: Accuracy by spatial relation axis. Depth-based relations requiring 3D reasoning are consistently harder than image-plane relations.

Relation Axis	GPT-5-mini	Claude Haiku 4.5	Qwen2.5-VL-7B
above/below	**93.8%**	**87.5%**	**65.4%**
left/right	78.1%	62.5%	61.2%
foreground/background	75.0%	75.0%	60.4%
front/behind	91.7%	45.8%	53.6%

Vertical relations (above/below) achieve the highest accuracy across all models, likely because they map directly to image y-coordinates without requiring depth inference. The front/behind axis shows the largest variance across models: GPT-5-mini achieves 91.7% while Claude Haiku 4.5 drops to 45.8%, suggesting significant differences in depth reasoning capabilities.

4.4 Hierarchical Breakdown

Table 4 provides the full hierarchical breakdown by task type, frame type, and relation axis. This fine-grained view reveals specific failure patterns that aggregate metrics obscure.

Table 4: Hierarchical accuracy breakdown. Cells show accuracy (%) for each model. Highlighted cells indicate accuracy below 50%.

Task / Frame	Relation	GPT-5	Claude	Qwen
Ego / Viewer	above/below	100.0	87.5	68.2
	fg/bg	75.0	75.0	62.1
	left/right	87.5	87.5	63.5
Ego / Obj-Obj	above/below	87.5	87.5	62.6
	front/behind	100.0	12.5	55.8
	left/right	75.0	62.5	58.9
Allo / Viewer	front/behind	100.0	62.5	54.2
	left/right	62.5	50.0	51.8
Allo / Obj-Obj	front/behind	75.0	62.5	52.1
	left/right	87.5	50.0	52.9

The hierarchical breakdown reveals a critical failure mode for Claude Haiku 4.5: egocentric object-to-object front/behind queries achieve only 12.5% accuracy—below chance level. This suggests a systematic misunderstanding of depth relations when reasoning about object pairs. In contrast, GPT-5-mini maintains robust performance across all conditions.

4.5 Summary of Findings

Our benchmark evaluation reveals three key patterns:

1. Consistent egocentric-allocentric gap. All models perform worse on allocentric tasks requiring perspective transformation. This gap ranges from 6.3 to 12.6 percentage points and persists across architectures, suggesting a fundamental limitation in current VLM training or architecture.

2. Depth reasoning is challenging. Front/behind relations show the highest variance across models and the largest accuracy drops. This indicates that inferring 3D spatial relations from 2D images remains difficult, even for state-of-the-art models.

3. Model-specific failure patterns. Different models exhibit distinct weaknesses: Claude Haiku 4.5 struggles with object-to-object relations and depth reasoning, while Qwen2.5-VL-7B shows more uniform (but lower overall) performance. These patterns suggest that targeted improvements rather than general scaling may be needed.

4.6 Mechanistic Analysis of Spatial Reasoning

Beyond aggregate accuracy metrics, we seek to understand *why* models succeed or fail at spatial reasoning. To this end, we conduct mechanistic analysis on Qwen2.5-VL-7B (8.29B parameters), extracting internal activations and attention patterns to identify neural correlates of correct spatial reasoning.

Method. We evaluate Qwen2.5-VL-7B on 50 COCO images, generating 3672 spatial queries that span all task types, frame types, and relation axes defined in our benchmark. Table 5 summarizes the results: the model achieves 63.9% accuracy on egocentric tasks but only 52.8% on allocentric tasks, confirming an 11.1 percentage point gap that indicates perspective transformation poses substantial difficulty.

Table 5: Qwen2.5-VL-7B spatial reasoning accuracy on 3672 queries across 50 COCO images.

Task Type	Accuracy (%)	Correct / Total
Egocentric	63.9	1477 / 2312
Allocentric	52.8	718 / 1360
Overall	**59.8**	**2195 / 3672**

To probe the internal mechanisms underlying these accuracy differences, we sample 33 cases—17 correct and 16 incorrect predictions—and extract two complementary signals. First, we compute *activation norms* (L2 norm of hidden states) at each layer, measuring representation strength. Second, we compute the *image attention ratio*, defined as the fraction of attention weights directed to image tokens versus text tokens, quantifying how strongly the model grounds its reasoning in visual content.

Results. Figure 2 presents our findings across three panels. Panel (A) shows activation norms across the model's layers, from early vision blocks through the language model. Notably, success and failure cases exhibit nearly identical activation patterns throughout the network, indicating that raw representation magnitude does not differentiate correct from incorrect predictions. This null result is informative: it suggests that failures do not stem from weak or degraded representations, but rather from how those representations are used.

The key differentiation emerges in attention patterns, shown in Panel (B). For egocentric tasks, a clear separation appears: successful predictions (green solid line) consistently maintain higher image attention than failures (red dashed line) across all language model layers, with the gap most pronounced at Layer 14. In contrast, allocentric tasks show no such separation—the success and failure lines interleave throughout the network, with successful predictions sometimes showing *lower* image attention than failures.

Panel (C) quantifies this divergence at Layer 14. For egocentric tasks, successful predictions exhibit 6.6 percentage points higher image attention than failures (50.3% vs. 43.8%). This pattern indicates that egocentric failures are fundamentally *grounding failures*—the model does not sufficiently attend to visual content when it answers incorrectly. For allocentric tasks, successful predictions show 1.6 percentage points *lower* attention to image tokens than failures (40.6% vs. 42.2%). This reversal indicates that allocentric failures are *reasoning failures*: the model attends to the image adequately

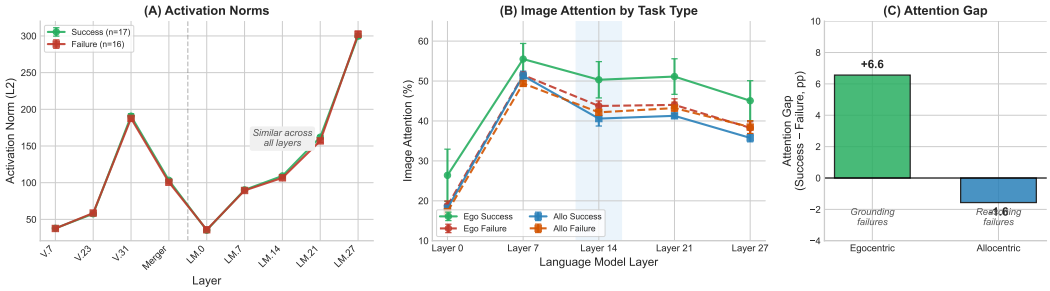

Figure 2: **Mechanistic analysis of spatial reasoning.** (A) Activation norms show no difference between success and failure cases across all layers, indicating that representation strength alone does not predict correctness. (B) Image attention ratio reveals task-specific patterns: egocentric success cases (green solid) consistently attend more to image tokens than failures (red dashed), while allocentric cases show no such separation. (C) At Layer 14, egocentric tasks exhibit a +6.6 percentage point attention advantage for successful predictions (grounding failures), whereas allocentric tasks show –1.6 percentage points (reasoning failures).

but cannot correctly perform the perspective transformation required to answer from an object's viewpoint.

Takeaways. These findings reveal that spatial reasoning failures arise from fundamentally different mechanisms depending on task type. The similarity in activation norms rules out representation degradation as a failure mode; instead, the divergence appears in how attention is allocated. For egocentric reasoning, success correlates with stronger visual grounding, suggesting that attention-focused training or architectural modifications could yield improvements. For allocentric reasoning, however, the absence of an attention deficit points to a deeper limitation in perspective-taking capabilities, which may require explicit modules or training objectives that teach reference frame transformation.

5 Conclusion